ORION

PART 07

Resilience & Emergency Systems

Disaster response, blackout mode, failover, self-healing infrastructure

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Resilience objective

Keep essential functions available when components, links, regions, or dependencies fail. Measure resilience rather than assuming it.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Failure lifecycle

Detect → classify → isolate → fail over/degrade → preserve essential functions → recover/reconcile → learn.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Fault tolerance

Redundant instances, multiple regions, multiple paths, replicated data, health checks, and controlled failover.

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Graceful degradation

Define a minimal essential service set and deliberately shed nonessential workloads during severe capacity loss.

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Blackout mode

A controlled lab exercise can disable selected simulated network paths and test emergency communication recovery without disrupting real systems.

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Retries and circuit breakers

Retry only transient errors. Use backoff, jitter, idempotency, and circuit breakers to prevent cascading failures.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Disaster recovery

Define RPO and RTO. Test backups by restoration. Maintain recovery runbooks and cross-region procedures.

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Resilience testing

Game days, dependency shutdowns, regional failover, queue overload, recovery drills, and post-test analysis.

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Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

Resilience & Emergency Systems should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.